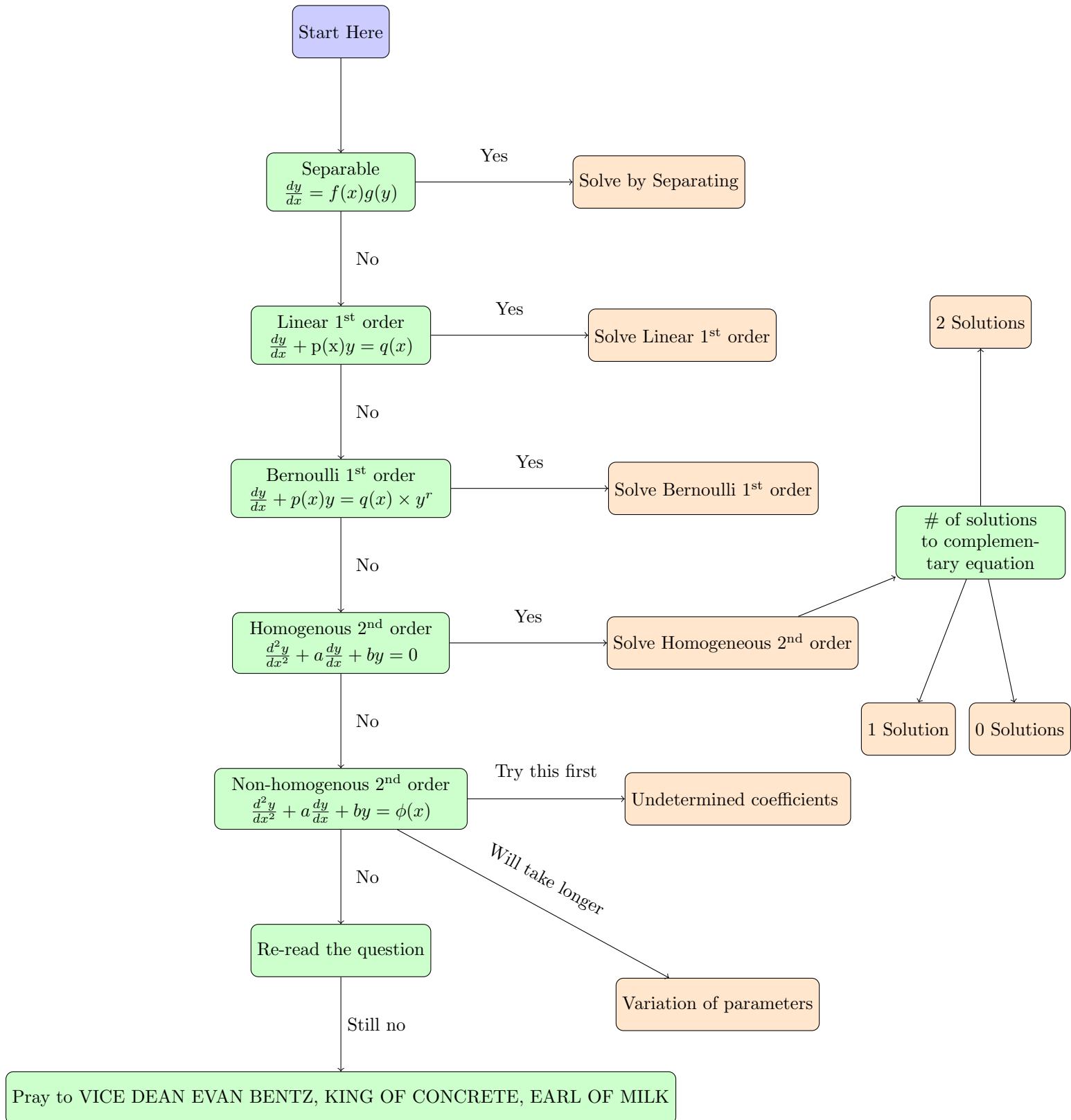


Flowchart inspired (first page mostly copied from) by: GOAT EngSci QILIN XUE!!!
 Derivations and examples mostly taken from: lecture or Paul's Online Math Notes



Separable Differential Equations

Background info

If you have an equation like $\frac{dy}{dx} = xy$, pretend $\frac{dy}{dx}$ is a fraction. Rearrange so that you get all the y terms on one side and all the x terms on the other. Then integrate with respect to the variable on each side. The answer will probably be explicit, but could be implicit; if so just leave it like that.

Example:

Problem:

$$y' = \frac{xy^3}{\sqrt{1+x^2}}, \quad y(0) = -1$$

Steps:

y' is $\frac{dy}{dx}$, then separate the variables:

$$\begin{aligned}\frac{dy}{dx} &= \frac{xy^3}{\sqrt{1+x^2}} \\ \frac{dy}{y^3} &= \frac{x}{\sqrt{1+x^2}} dx\end{aligned}$$

Integrate! (u-sub for the right side)

$$\begin{aligned}\int \frac{dy}{y^3} &= \int \frac{x}{\sqrt{1+x^2}} dx \\ -\frac{1}{2y^2} &= \sqrt{1+x^2} + C\end{aligned}$$

Apply initial condition to find C :

$$\begin{aligned}-\frac{1}{2(-1)^2} &= \sqrt{1+0^2} + C \\ -\frac{1}{2} &= 1 + C \\ C &= -\frac{3}{2}\end{aligned}$$

Find explicit solution for $y(x)$

$$\begin{aligned}-\frac{1}{2y^2} &= \sqrt{1+x^2} - \frac{3}{2} \\ \frac{1}{y^2} &= 3 - 2\sqrt{1+x^2} \\ y^2 &= \frac{1}{3 - 2\sqrt{1+x^2}} \\ y &= \pm \sqrt{\frac{1}{3 - 2\sqrt{1+x^2}}}\end{aligned}$$

Note that given the initial condition, we only take the negative root. Also note that the domain of x is restricted.

Sample Problem: (Stewart 9.3.18)

$$\frac{dL}{dt} = kL^2 \ln t, \quad L(1) = -1$$

Linear 1st Order Differential Equations

Background info:

The equation has to be in this form $\frac{dy}{dx} + p(x)y = q(x)$. Assume there is some function $\mu(x)$ (the integrating factor) has the property $\mu(x)p(x) = \mu'(x)$ (it later turns out that $\mu(x) = e^{\int p(x)dx}$).

Multiply through by $\mu(x)$ to get $\mu(x)\frac{dy}{dx} + \mu'(x)y = \mu(x)q(x)$. The left hand side of this equation is just the product rule of $(\mu(x)y(x))'$, so the equation becomes $(\mu(x)y(x))' = \mu(x)q(x)$. Integrate both sides **and keep the constant of integration from this step onwards**.

Isolate for $y(t)$ to get the solution to the equation:

$$y(t) = \frac{\int \mu(t)q(t)dt - c}{\mu(t)}$$

where $\mu(x) = e^{\int p(x)dx}$. Note the sign of the constant doesn't matter.

Example:

Problem:

$$\frac{dv}{dt} = 9.8 - 0.196v, \quad v(0) = 48$$

Steps:

$$p(t) = -0.196 \text{ and } q(t) = 9.8 \quad \mu(t) = e^{\int p(t)dt} = e^{\int -0.196 dt} = e^{-0.196t}$$

Multiply through to get $\frac{d}{dt}(e^{-0.196t}v) = 9.8e^{-0.196t}$

Integrate both sides and divide out to find explicit solution for v: $v(t) = 50 + Ce^{-0.196t}$

Use initial condition to find that $C = -2$, so final answer becomes: $v(t) = 50 - 2e^{-0.196t}$

Bernoulli 1st Order Differential Equations

Background info

Given a differential equation in this form $\frac{dy}{dx} + p(x)y = q(x)y^n$, where $n \neq 0$ or 1 (because that's just a linear 1st order differential equation (see above)). Divide through by y^n and make the substitution $v = y^{1-n}$. Note $\frac{dv}{dx} = (1-n)y^{-n}\frac{dy}{dx}$, so the equation can be rearranged to

$$\frac{v'(x)}{1-n} + p(x)v = q(x)$$

which is just a linear differential equation (with a different variable!!).

Example:

Problem:

$$y' + \frac{4}{x}y = x^3y^2, \quad y(2) = -1$$

Steps:

Let $v = y^{1-2} = y^{-1}$ and thus $y' = -v^{-2}v'$

Substitute to get $-v^{-2}v' + \frac{4}{x}v^{-1} = x^3v^{-2}$

Divide through to get a linear first order equation which has an integrating factor of $e^{\int -\frac{4}{x}dx} = e^{-4\ln x} = x^{-4}$

Multiply through to get $x^{-4}v' = \int -x^{-1}dx$

Integrate both sides and divide out to find explicit solution for v: $v(x) = x^4(c - \ln x)$

We need an explicit solution for y and the value of c, so we'll find both. The equation for y becomes $y^{-1} = x^4(c - \ln x)$. Applying the initial condition, we get $c = \ln 2 - \frac{1}{16}$

Homogeneous 2nd Order Differential Equations

Background info:

The equation should be in the form $ay'' + by' + cy = 0$.

The **Principle of Superposition** states that if $y_1(x)$ and $y_2(x)$ are two linearly independent “nice enough” solutions (the Wronskian doesn’t disappear or something like that. Irrelevant for 194), then the general solution to the differential equation is $y(x) = c_1y_1(x) + c_2y_2(x)$, where c_1, c_2 are some constants.

Complementary equation

Background info:

Set the polynomial to zero and make it a homogeneous equation. **Do NOT find the coefficients**

Homogeneous 2nd Order Differential Equations: One Solution

Background:

If there’s only one real root, then if we try the method with two roots we get $y_1 = y_2$, which doesn’t really work. Magic happens and then the solution becomes:

$$y(x) = c_1e^{-bx/2a} + c_2xe^{-bx/2a}$$

Sample Problem:

Solve the differential equation $y'' - 4y' + 4y = 0$.

Homogeneous 2nd Order Differential Equations: Zero Solutions

Background info:

If the roots are not real, use **Euler’s formula**: $e^{i\theta} = \cos \theta + i \sin \theta$ (note you can (will) use the form where θ is negative, in that case, remember the even/odd behavior of sin/cos). The following equations use $\lambda = \frac{-b}{2a}$ and $\mu = \sqrt{b^2 - 4ac}$, where μ is just the real part (multiply by i).

$$y_1(x) = e^{(\lambda + \mu i)x} \quad \text{and} \quad y_2(x) = e^{(\lambda - \mu i)x}$$

$$y_1(x) = e^{\lambda x} e^{\mu i x} = e^{\lambda x} (\cos(\mu x) + i \sin(\mu x))$$

$$y_2(x) = e^{\lambda x} e^{-\mu i x} = e^{\lambda x} (\cos(\mu x) - i \sin(\mu x))$$

Add/subtract the two solutions together to get both solutions, and then you get:

$$y(x) = c_1 e^{\lambda x} \cos(\mu x) + c_2 e^{\lambda x} \sin(\mu x)$$

Homogeneous 2nd Order Differential Equations: Two Solutions

Background info:

A likely candidate for $y(x)$ is something along the lines of e^{rx} , where r is some constant. If that was the case, then by taking the derivative twice, the equation becomes $e^{rx}(ar^2 + br + c) = 0$. Since $e^{\text{something}}$ is

never 0, the polynomial has to be zero at some point. Factor it to get the possible values of r , and then if they're real, plug in and you're done.

Method of undetermined coefficients

Background info:

$$ay'' + by' + cy = g(x)$$

Similarlyish to solving for homogenous equations, the solution is going to be something along the lines of the solution to the complementary equation + the actual solution. **1. Guess y_p based on $g(x)$:**

- **Case 1:** $g(x)$ is a polynomial. Guess: $y_p = Ax^n + Bx^{n-1} + \dots + Cx + D$. Even if there are not all the polynomial terms in the original $g(x)$, there still has to be all of them in the equation.
- **Case 2:** $g(x)$ is exponential (e^{kx}). Guess: $y_p = Ae^{kx}$.
- **Case 3:** $g(x)$ is sine or cosine ($\sin(kx)$ or $\cos(kx)$). Guess: $y_p = A\cos(kx) + B\sin(kx)$, or something easier if it clearly cancels.

Notes:

- If $g(x)$ combines functions, combine guesses.
- If any y_p term solves the homogeneous equation, multiply y_p by x (or x^2) until no term is a homogeneous solution.

Substitute y_p and its derivatives into the differential equation, then equate coefficients to find the undetermined coefficients.

Variation of parameters

Background info:

Same idea as undetermined coefficients. Start by assuming a particular solution of the form

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

where $u_1(x)$ and $u_2(x)$ are functions to be determined. u is some made up thing so we can force it to follow $u'_1y_1 + u'_2y_2 = 0$ Plugging and chugging, we get.

$$y'_p = u'_1y_1 + u'_2y_2 + u_1y'_1 + u_2y'_2$$

$$y''_p = u'_1y'_1 + u'_2y'_2 + u_1y''_1 + u_2y''_2$$

$$a(u'_1y'_1 + u'_2y'_2 + u_1y''_1 + u_2y''_2) + b(u_1y'_1 + u_2y'_2) + c(u_1y_1 + u_2y_2) = g(x)$$

$$u_1(ay''_1 + by'_1 + cy_1) + u_2(ay''_2 + by'_2 + cy_2) + a(u'_1y'_1 + u'_2y'_2) = g(x)$$

Since $ay''_1 + by'_1 + cy_1 = 0$ and $ay''_2 + by'_2 + cy_2 = 0$ (solutions to complementary equation), we can simplify to $a(u'_1y'_1 + u'_2y'_2) = g(x)$ Integrate and substitute as necessary.

Relevant exam question

(a)

Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of

$$y'' + ay' + by = 0$$

where a and b are constants. By proving that $W'(x) = -aW(x)$, show that either $W(x) = 0$ for all x or $W(x) \neq 0$ for any x .

Proof.

1. **Derivative of the Wronskian:** The Wronskian is defined as $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. Differentiating with respect to x , we get:

$$\begin{aligned} W'(x) &= y_1'(x)y_2'(x) + y_1(x)y_2''(x) - y_2'(x)y_1'(x) - y_2(x)y_1''(x) \\ &= y_1(x)y_2''(x) - y_2(x)y_1''(x) \end{aligned}$$

2. **Using the differential equation:** Since y_1 and y_2 are solutions to the differential equation $y'' + ay' + by = 0$, we have:

$$\begin{aligned} y_1''(x) &= -ay_1'(x) - by_1(x) \\ y_2''(x) &= -ay_2'(x) - by_2(x) \end{aligned}$$

Substituting these into the expression for $W'(x)$:

$$\begin{aligned} W'(x) &= y_1(x)[-ay_2'(x) - by_2(x)] - y_2(x)[-ay_1'(x) - by_1(x)] \\ &= -ay_1(x)y_2'(x) - by_1(x)y_2(x) + ay_2(x)y_1'(x) + by_2(x)y_1(x) \\ &= -a[y_1(x)y_2'(x) - y_2(x)y_1'(x)] \\ &= -aW(x) \end{aligned}$$

3. **Solving the differential equation for $W(x)$:** The differential equation $W'(x) = -aW(x)$ has the solution: $W(x) = W(0)e^{-ax}$
4. **Conclusion:** Since $e^{-ax} \neq 0$ for all x , $W(x) = 0$ if and only if $W(0) = 0$. Therefore, either $W(x) = 0$ for all x , or $W(x) \neq 0$ for any x .

□

(b)

Let $P(x)$ and $Q(x)$ be given, continuous functions. Let $y_1(x)$ and $y_2(x)$ be two solutions of

$$y'' + P(x)y' + Q(x)y = 0$$

such that their Wronskian $W(x)$ never vanishes. Show that between two consecutive zeros of $y_1(x)$, there is one and only one zero of $y_2(x)$.

Proof.

1. **Rolle's Theorem:** If a function $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b) = 0$, then there exists at least one $c \in (a, b)$ such that $f'(c) = 0$.
2. **Applying Rolle's Theorem to $y_1(x)$:** Let a and b be consecutive zeros of $y_1(x)$, so $y_1(a) = y_1(b) = 0$. As a solution to a second-order linear differential equation with continuous coefficients, $y_1(x)$ is continuous and differentiable. By Rolle's Theorem, there exists $c \in (a, b)$ such that $y_1'(c) = 0$.

3. **Using the Wronskian:** At $x = c$, the Wronskian is:

$$\begin{aligned} W(c) &= y_1(c)y_2'(c) - y_2(c)y_1'(c) \\ &= y_1(c)y_2'(c) \quad (\text{since } y_1'(c) = 0) \end{aligned}$$

Since $W(c) \neq 0$ and $y_1(c) = 0$, we must have $y_2'(c) \neq 0$.

4. **Applying Rolle's Theorem to $y_2(x)$:** If $y_2(a) = y_2(b) = 0$, then by Rolle's Theorem, there would exist $d \in (a, b)$ such that $y_2'(d) = 0$. But $y_2'(c) \neq 0$ and $c \in (a, b)$, so $y_2(x)$ cannot have two zeros between a and b .
5. **Conclusion:** We have shown that $y_2(x)$ cannot have two zeros between consecutive zeros of $y_1(x)$. Since the Wronskian never vanishes, $y_1(x)$ and $y_2(x)$ cannot have a common zero. Therefore, there is one and only one zero of $y_2(x)$ between two consecutive zeros of $y_1(x)$.

□